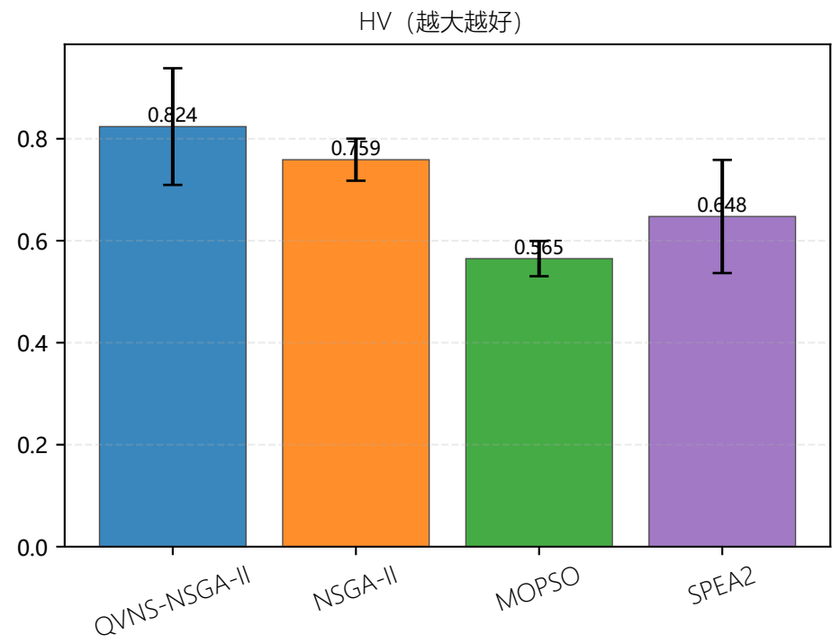
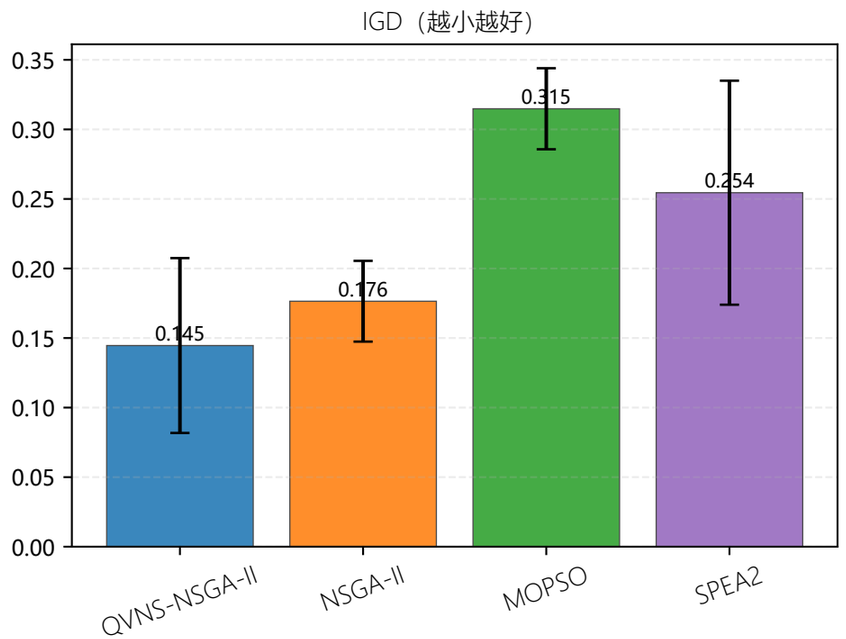
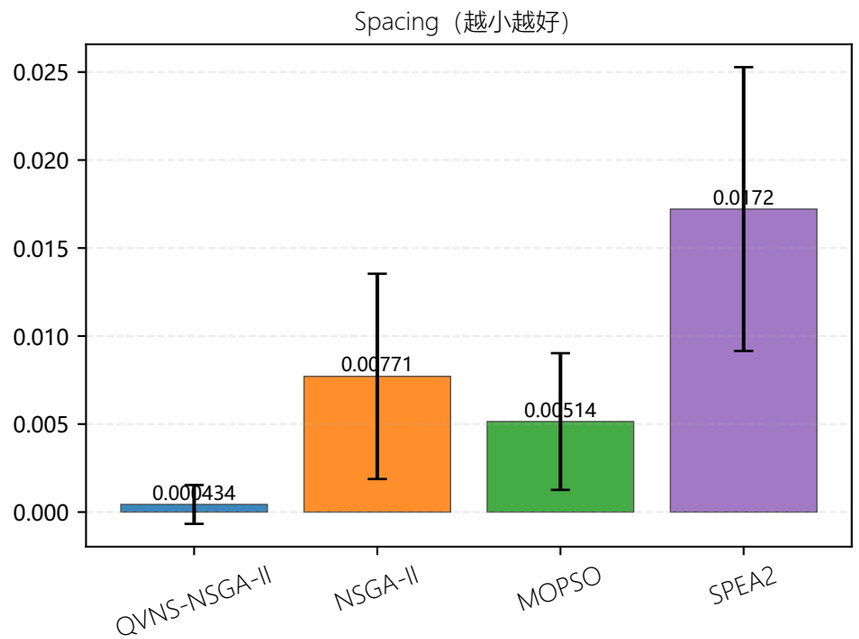
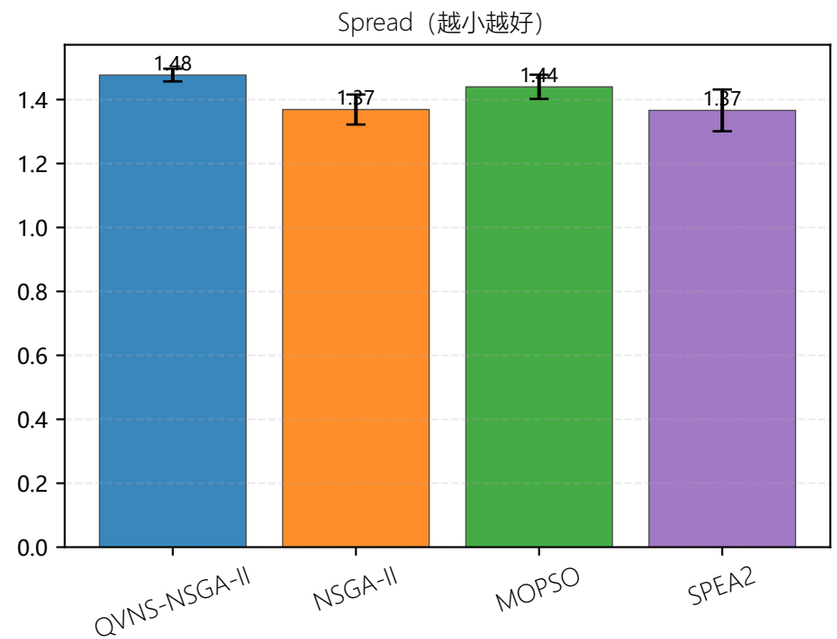
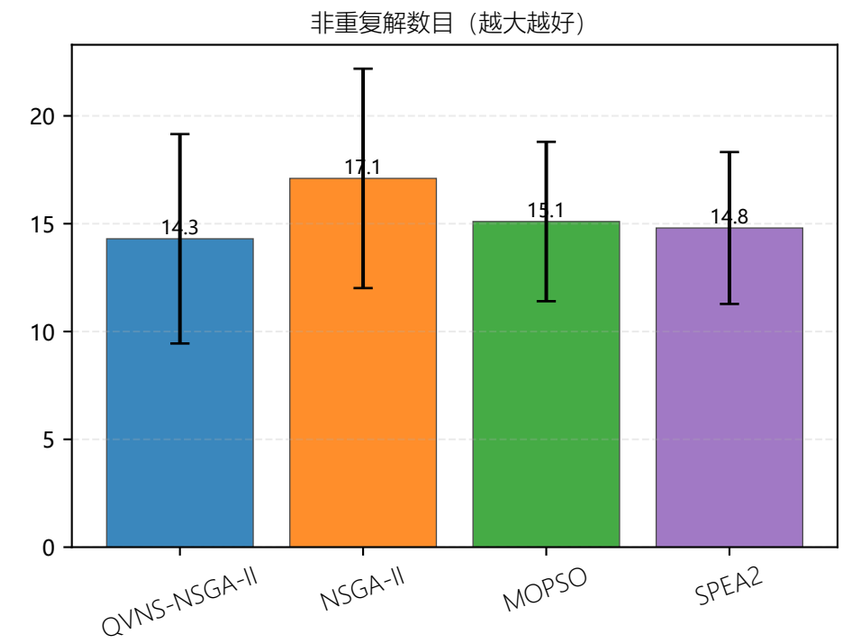
四种算法性能指标对比

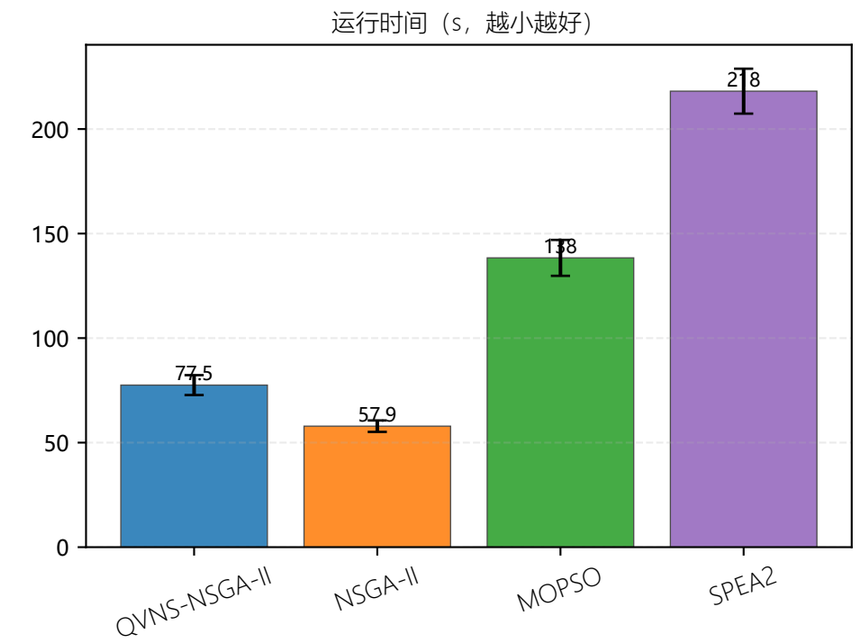

HV（越大越好）
0.824
0.759
0.565
0.648
0.0
0.2
0.4
0.6
0.8
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2


IGD（越小越好）
0.145
0.176
0.315
0.254
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2


Spacing（越小越好）
0.000434
0.00771
0.00514
0.0172
0.000
0.005
0.010
0.015
0.020
0.025
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2


Spread（越小越好）
1.48
1.37
1.44
1.37
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2


非重复解数目（越大越好）
14.3
17.1
15.1
14.8
0
5
10
15
20
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2


运行时间（s，越小越好）
77.5
57.9
138
218
0
50
100
150
200
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2